ST2195 COURSEWORK

Name : Stacia Tan Rui Shi

Student ID : 240509602

Table of Contents

Part 1 : Markov Chain Monte Carlo Algorithm ***3***

Random Walk Metropolis-Hastings Algorithm **3**

R **Convergence Diagnostic** **4**

Part 2 : USA Commercial Flight Data Analysis ***5***

Introduction **5**

Part 2a – What are the best times and days of the week to minimize delays each year? **5**

Data analysis using R : 6

Yearly trend (2002-2006) : 7

Part 2b - Evaluate whether older planes suffer more delays on a year-to-year basis. **7**

Assessment of Plane Data 7

Overview : 8

Part 2c - Logistic Regression Using Machine Learning **9**

Synopsis 9

Data Analysis for Logistic Regression Coefficient 10

List of Figures

Figure 1: Random Walk Histogram 3

Figure 2 : Line Graph - Convergence Diagnostic 4

Figure 3 :Delay Classification 5

Figure 4: Flight Timing Classification 5

Figure 5 : Bar Graph - Percentage of Flight Delays By Month 6

Figure 6: Bar Graph - Percentage of Flight Delays By Day 6

Figure 7 : Bar Graph - Percentage of Flight Delays by Time of Day 6

Figure 8 : Summary of Findings 6

Figure 9 : Heat Map - Day of Week x Hour 6

Figure 10 : Line Plot - 5-Year Trend : Delay Risk by Time of Day (2002-2006) 7

Figure 11 : Line Plot - 5-Year Trend : Delay Risk by Day of the Week 7

Figure 12: Scatter Plot - Does Plane Age Affect Average Flight Delay? 8

Figure 13: Scatter Plot - Does Plane Age Affect Average Flight Delay (Yearly)? 8

Figure 14 : Model Accuracy Percentage By Year (ran in R) 9

Figure 15: Logistic Regression Coefficients 10

Figure 16: Key Drivers of Flight Diversions (2002-2006) 10

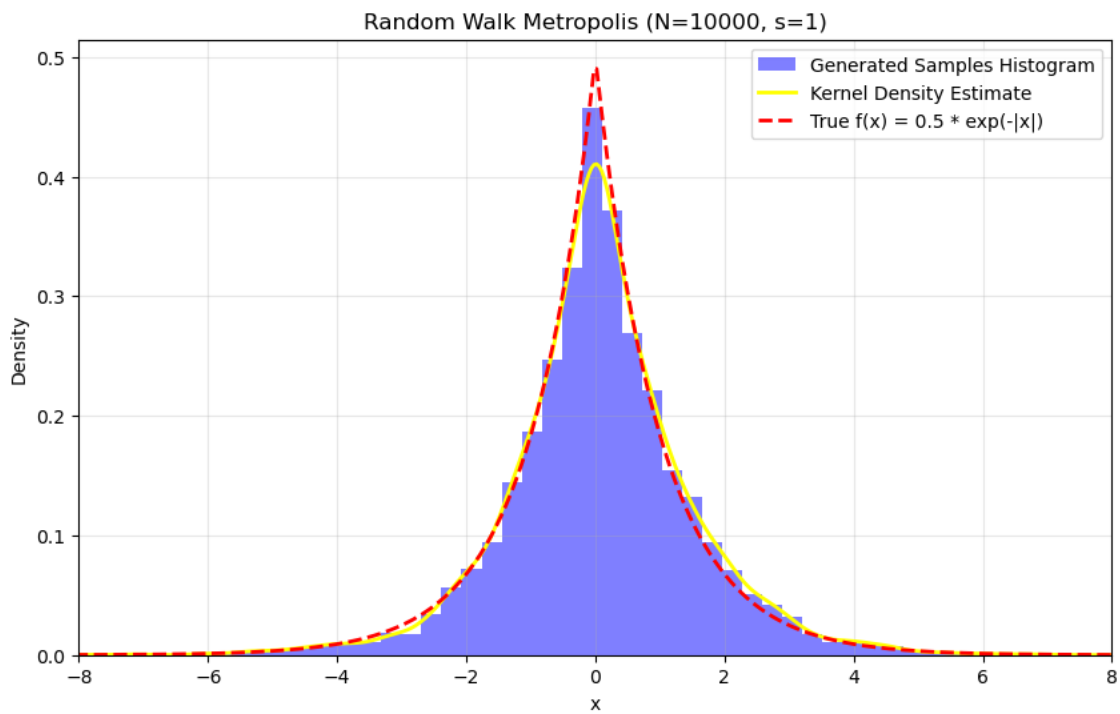
Part 1 : Markov Chain Monte Carlo Algorithm

Random Walk Metropolis-Hastings Algorithm

The Monte Carlo method measures estimation of complex distribution through repeated random sampling and Markov Chain is a stochastic process is the probability of the next step referring solely on the current state. The Markov Chain Monte Carlo Algorithm- Random Walk Metropolis generates a random “walk” of N iterations by proposing, accepting and/or rejecting steps based on step size (s).

Using Python, as we implement Random Walk Metropolis using N = 10000 and s = 1, we attained :

Figure 1: Random Walk Histogram



Where we calculated that Sample Mean is 0.0723 and Sample Standard Deviation : 1.4269

The algorithm relies on pseudo-random number generation. Hence, the sample paths for R and Python differ slightly. However, both successfully converged to near-identical means and standard deviations, confirming the algorithm's accuracy.

$\hat{R}$ Convergence Diagnostic

$\hat{R}$ is used to confirm if Markov chains have successfully converged. The standard threshold for convergence is $\hat{R} < 1.05$.

Conditions for this question states that : N = 2000, s = 0.001, J =4

Using R : We achieve a $\hat{R}$ value of 126.108 when s=0.001

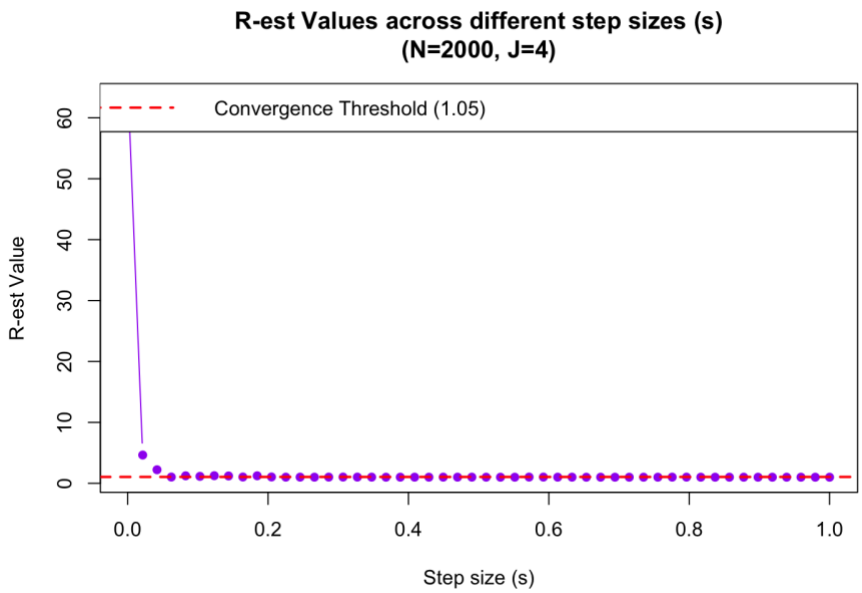


Figure 2 : Line Graph - Convergence Diagnostic

For a step size of s = 0.001, the calculated $\hat{R}$ value is 63.13 - significantly higher than the 1.05 threshold. Indicating that the chains failed to converge as the step size of 0.001 is too small. When the step size increases past approximately 0.25, the $\hat{R}$ value consistently drops below the 1.05 threshold, demonstrating successful convergence of the Markov chains.

Part 2 : USA Commercial Flight Data Analysis

Introduction

Using data analysis for 5 years of flight data- we intend to find the best day of the week, hour and month to depart to minimize delays. This is crucial as some travellers are booking flights for work, weddings or connecting flights and cannot afford to be late.

To process millions of flight data- SQLite database was used to handle the large volume of data.

- The raw CSV files for each year were read and appended into ontime database table
- Secondary time planes was extracted from plane-data.csv for q2b

Data Preparation and Methodology

1. Data cleaning – filtering invalid, cancelled and diverted flights as we are only focusing on delayed flights
2. Missing values – any records with NA values were dropped, it would be illogical to replace them with average or previous data entry as our objective is to measure actual ontime and delayed flights. Doing so would artificially inflate the delay data with synthetic records, leading to inaccurate percentages. Dropping NA values ensures our findings are based entirely on observed, real-world flight data.
3. Standardizing Time – midnight departures were counted as 24** instead of 00**, so we had to convert it to ensure accuracy of 24hour cycle
4. Plane age – when calculating we removed any negative and missing age value

Delay classification

Classification	Delay (Minutes)
Ontime	≤15
Minor	< 60
Moderate	< 120
Severe	>120

Figure 3 :Delay Classification

Flight timing classification

Classification	Scheduled Departure Time Range
Red-eye	0000 to 0559
Morning	0600 to 1159
Afternoon	1200 to 1759
Night	1800 to 2359

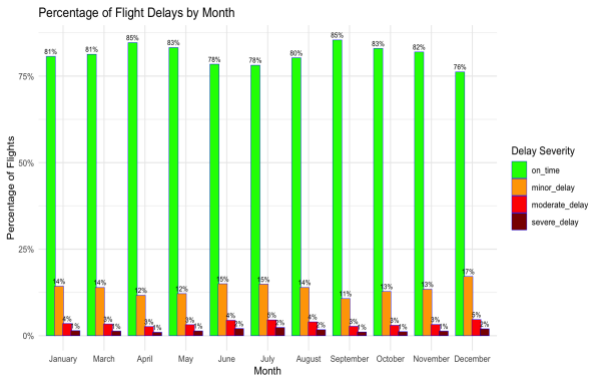
Figure 4: Flight Timing Classification

Part 2a – What are the best times and days of the week to minimize delays each year?

To categorize Arrival Delay- delay classification was used. 'Arrival Delay' was used as the variable to calculate the average delay risk for each specific time slot. By visualizing the data in this matrix, consumers can easily identify historical delay patterns and optimize their flight booking decisions to avoid peak delay periods. The data was then arranged across three dimensions – Month, Day of the Week and Time of Day.

Data analysis using R :

Figure 5 : Bar Graph - Percentage of Flight Delays By Month



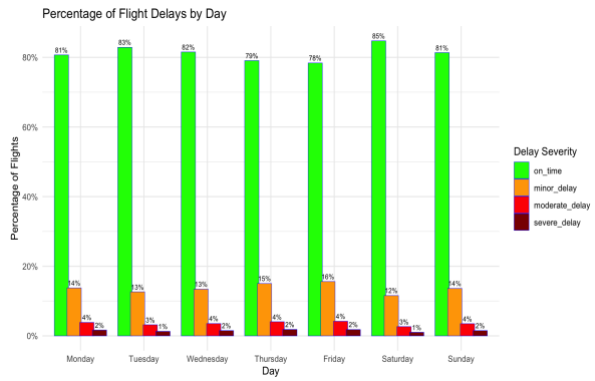
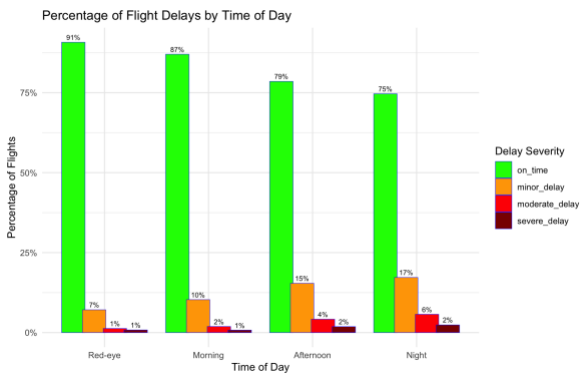


Figure 6: Bar Graph - Percentage of Flight Delays By Day

Figure 7 : Bar Graph - Percentage of Flight Delays by Time of Day

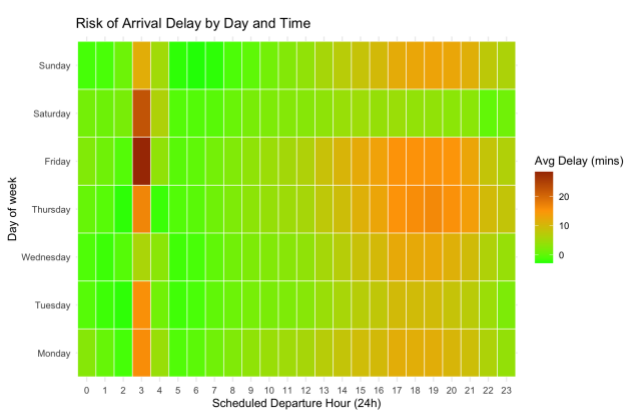


From the 3 graphs, we can infer that :

Best _ To Fly :	Worst _ To Fly :
Month : September	Month : December
Day of The Week : Saturday	Day of The Week : Friday
Time of The Day : Red-eye	Time of The Day : Night

Figure 8 : Summary of Findings

Figure 9 : Heat Map - Day of Week x Hour



To provide a more insightful optimal departure day and time for travelers- a heatmap was generated cross-referencing 'Scheduled Departure Hour' with 'Day of the Week'.

Yearly trend (2002-2006) :

Figure 10 : Line Plot - 5-Year Trend : Delay Risk by Time of Day (2002-2006)

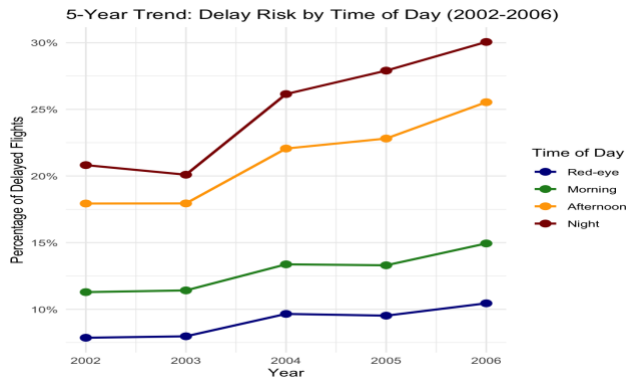
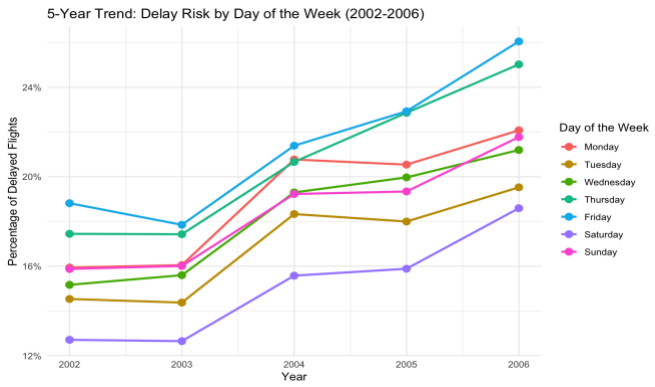


Figure 11 : Line Plot - 5-Year Trend : Delay Risk by Day of the Week



The 5-year trend proves that our data is consistent yearly.

To minimize the risk of delays, consumers should book red-eye flights and on Saturday

Part 2b - Evaluate whether older planes suffer more delays on a year-to-year basis.

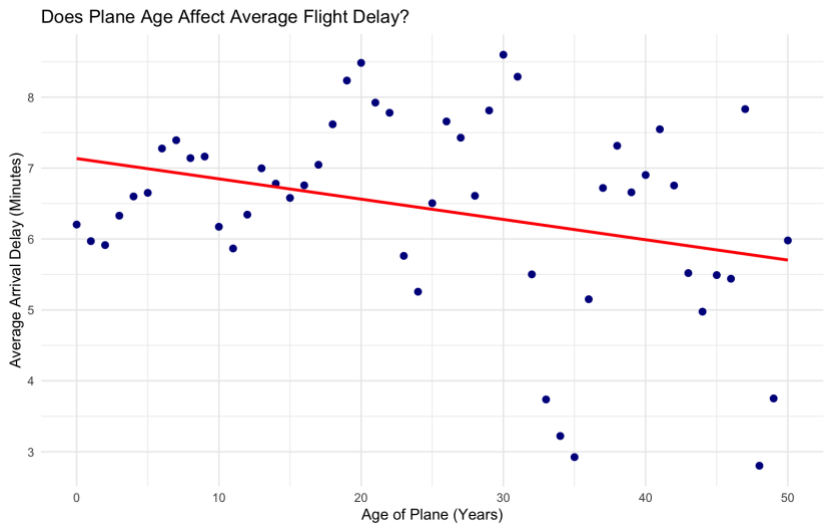
Assessment of Plane Data

To evaluate the impact of aircraft age on performance, the primary flight dataset was cross-referenced with a secondary database containing aircraft manufacturing details.

- Data Integration: The two datasets were merged using the aircraft's unique 'Tail Number' as the primary key.
- Data Cleaning : Invalid records (e.g. missing manufacturing years or negative plane ages) were filtered out. The data was then aggregated to find the mean arrival delay for each specific age group before applying a linear regression model to identify overall trends.
- Feature Engineering: A new 'Plane Age' variable was calculated by subtracting the aircraft's manufacturing year from the year of the flight.

Overview :

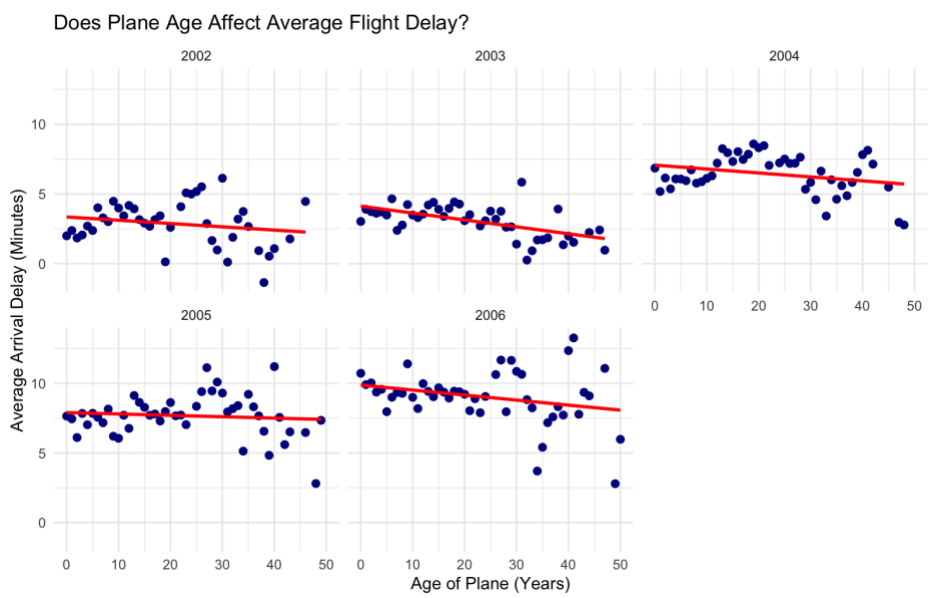
Figure 12: Scatter Plot - Does Plane Age Affect Average Flight Delay?



The analysis reveals that the assumption regarding older aircraft suffering more delays is fundamentally incorrect. Linear regression indicates slightly negative correlation across 2002-2006. Older planes do not experience higher average delays.

A comprehensive yearly analysis shows that older aircraft do not suffer more delays

Figure 13: Scatter Plot - Does Plane Age Affect Average Flight Delay (Yearly)?



Part 2c - Logistic Regression Using Machine Learning

Synopsis

To predict the probability of a flight being diverted- mlr3 (machine learning pipeline) was constructed in R. The dataset contains millions of records, to prevent my laptop from crashing- a random sample of 100,000 flights was extracted per year (2002-2006).

Statistically, achieving a 95% confidence level (the standard default in most scientific research) with a strict 1% margin of error requires a sample size of approximately 9,600. Therefore, our sample of 100,000 far exceeds the required threshold, ensuring the model's coefficients are highly representative of the full 6-million flight dataset.

A logistic regression learner was utilized to predict diversion probabilities, calculating the log-odds coefficients using core flight metrics - scheduled times, distance, and coordinates and categorical variables (airlines and days of the week).

The data was separated into 80% train set, 20% test set.

To interpret the model's findings, the coefficients were extracted and visualized. Due to the high dimensionality created by categorical dummy variables (such as individual airline performance), two visual models were generated: a comprehensive facet grid containing all variables, and a refined summary chart isolating the core, actionable flight metrics to track their impact evolution over the five-year timeframe.

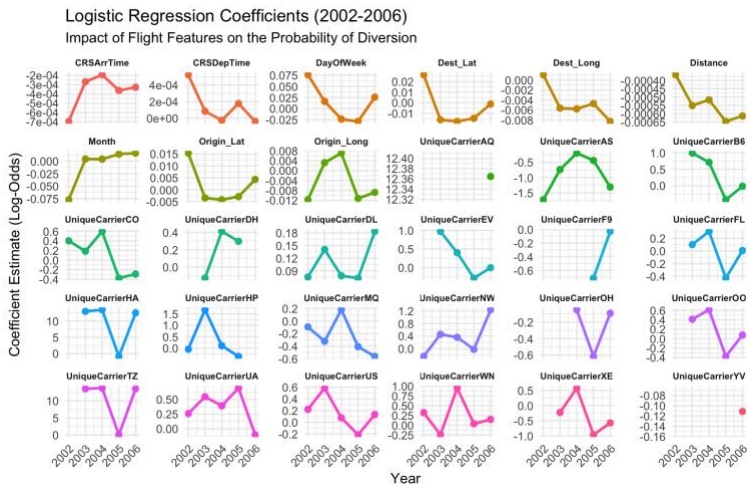
```
+ }
[1] "Processing Year: 2002"
[1] "Year 2002 Model Accuracy: 0.9984"
[1] "Processing Year: 2003"
[1] "Year 2003 Model Accuracy: 0.9986"
[1] "Processing Year: 2004"
[1] "Year 2004 Model Accuracy: 0.9981"
[1] "Processing Year: 2005"
[1] "Year 2005 Model Accuracy: 0.9978"
[1] "Processing Year: 2006"
[1] "Year 2006 Model Accuracy: 0.9976"
```

Figure 14 : Model Accuracy Percentage By Year (ran in R)

The logistic regression models achieved a prediction accuracy of approximately 99.8% across all five years. However, accuracy alone can be misleading in this context due to severe class imbalance. Flight diversions represent a remarkably small fraction of total flights, a baseline model predicting '0' - Not Diverted, would yield a similarly high accuracy score. Despite this, the extraction of the log-odds coefficients remains highly valuable, as it successfully separates the directional impact of our independent variables (i.e. distance and coordinates) on the underlying probability of a diversion occurring.

Data Analysis for Logistic Regression Coefficient

Figure 15: Logistic Regression Coefficients



The full coefficient plot shows that airline carrier variables exhibit the largest variation in log-odds across years, indicating that operational differences between carriers are a key driver of flight diversion probability. In contrast, temporal variables such as scheduled departure and arrival times have consistently small coefficients, suggesting a limited direct impact. Geographic variables (latitude and longitude) show fluctuating effects across years, implying that their influence is less stable and likely context-dependent.

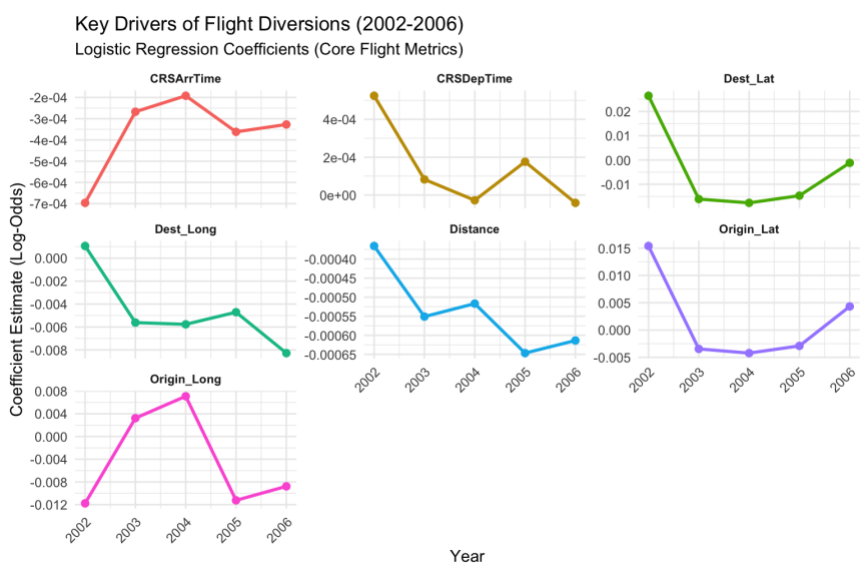



Figure 16: Key Drivers of Flight Diversions (2002-2006)